

Max Option Contract Solutions

Parts A and B Step-by-Step

Part A: Fair Value Formula at $t = 0$

Given: $V(T) = \max\{S_1(T), S_2(T)\}$, $S_1(0) = S_2(0) = 100$, independent processes ($\rho = 0$), and equal volatility ($\sigma_1 = \sigma_2 = \sigma$).

Step 1: Decompose the Payoff

Rewrite the terminal payoff as holding one stock plus an exchange option:

$$V(T) = S_1(T) + \max(S_2(T) - S_1(T), 0)$$

Step 2: Apply Margrabe's Formula

The general value for exchanging one asset for another is:

$$V(t, S_1, S_2) = S_1 N(d_1) + S_2 N(d_2)$$

where $N(\cdot)$ is the cumulative standard normal distribution.

Step 3: Calculate Combined Volatility

The combined volatility for independent stocks is:

$$\hat{\sigma} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \sqrt{\sigma^2 + \sigma^2} = \sigma\sqrt{2}$$

Step 4: Evaluate at $t = 0$

Substitute $S_1(0) = S_2(0) = 100$. Since $S_1/S_2 = 1$, $\ln(1) = 0$:

$$d_1 = \frac{\ln(S_1/S_2) + \frac{1}{2}\hat{\sigma}^2 T}{\hat{\sigma}\sqrt{T}} = \frac{0 + \frac{1}{2}(\sigma\sqrt{2})^2 T}{\sigma\sqrt{2}T} = \frac{\sigma^2 T}{\sigma\sqrt{2}T} = \frac{\sigma\sqrt{T}}{\sqrt{2}}$$

Due to symmetry, $d_2 = d_1$.

Step 5: Final Formula

Final Formula for Part A

$$V(0) = 100N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) + 100N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right) = 200N\left(\frac{\sigma\sqrt{T}}{\sqrt{2}}\right)$$

Part B: Partial Differential Equation (PDE)

Step 1: Define Asset Dynamics

Assume Geometric Brownian Motion (GBM):

$$dS_1 = \mu_1 S_1 dt + \sigma S_1 dW_1$$

$$dS_2 = \mu_2 S_2 dt + \sigma S_2 dW_2$$

Independent stocks mean $dW_1 dW_2 = 0$.

Step 2: Apply 2D Ito's Lemma

Expand the differential $dV(t, S_1, S_2)$:

$$dV = \left(\frac{\partial V}{\partial t} + \mu_1 S_1 \frac{\partial V}{\partial S_1} + \mu_2 S_2 \frac{\partial V}{\partial S_2} + \frac{1}{2} \sigma^2 S_1^2 \frac{\partial^2 V}{\partial S_1^2} + \frac{1}{2} \sigma^2 S_2^2 \frac{\partial^2 V}{\partial S_2^2} \right) dt + \sigma S_1 \frac{\partial V}{\partial S_1} dW_1 + \sigma S_2 \frac{\partial V}{\partial S_2} dW_2$$

Step 3: Construct a Riskless Portfolio

Create a delta-hedged portfolio $\Pi = V - \Delta_1 S_1 - \Delta_2 S_2$.

Choose hedge ratios to eliminate dW_1 and dW_2 :

$$\Delta_1 = \frac{\partial V}{\partial S_1}, \quad \Delta_2 = \frac{\partial V}{\partial S_2}$$

The deterministic change in the portfolio is:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_1^2 \frac{\partial^2 V}{\partial S_1^2} + \frac{1}{2} \sigma^2 S_2^2 \frac{\partial^2 V}{\partial S_2^2} \right) dt$$

Step 4: Apply No-Arbitrage Condition

The riskless portfolio earns the risk-free rate r :

$$d\Pi = r\Pi dt = r \left(V - S_1 \frac{\partial V}{\partial S_1} - S_2 \frac{\partial V}{\partial S_2} \right) dt$$

Step 5: Formulate Final PDE

Equating the expressions for $d\Pi$ and rearranging yields the PDE:

Final PDE and Terminal Condition

$$\frac{\partial V}{\partial t} + r S_1 \frac{\partial V}{\partial S_1} + r S_2 \frac{\partial V}{\partial S_2} + \frac{1}{2} \sigma^2 S_1^2 \frac{\partial^2 V}{\partial S_1^2} + \frac{1}{2} \sigma^2 S_2^2 \frac{\partial^2 V}{\partial S_2^2} - rV = 0$$

Terminal Condition: $V(T, S_1, S_2) = \max(S_1, S_2)$